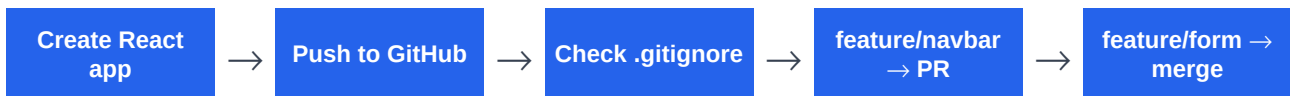


Exercise 5 — Upload React Project to GitHub

Step-by-step guide for first-time users (Windows, Git Bash).

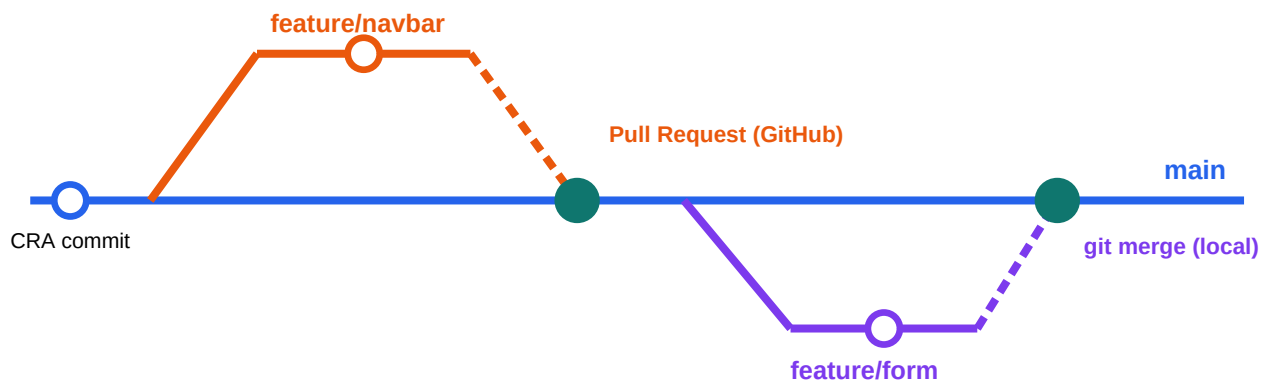
What We Will Do



Words You Must Know

Word	Simple meaning
node_modules	Huge folder of installed packages (thousands of files). Never upload it
.gitignore	A file listing things Git should ignore (not track, not upload)
Pull Request (PR)	A request on GitHub to merge one branch into another — done with buttons
git pull	Download the latest changes from GitHub to your computer
feature/navbar	Just a branch name. The "/" is only for neat naming

The Branch Plan



Before you start: finish Exercise 4 first (Git installed, name/email configured, GitHub token ready).

Green \$ lines = what you type (don't type the \$). White lines = output.

Part 1 — Create a New React App

AIM

To create a React app, upload it to GitHub, use .gitignore, create branches, and practise merging and pull requests.

Step 1 — Create the app (takes 2–5 minutes, needs internet)

```
● ● ● Git Bash

$ npx create-react-app myapp
Creating a new React app in C:\Users\You\myapp.
Installing packages. This might take a couple of minutes.
...
Initialized a git repository.
Success! Created myapp at C:\Users\You\myapp
Happy hacking!
```

Step 2 — Run it to check it works

```
● ● ● Git Bash

$ cd myapp
$ npm start
Compiled successfully!
You can now view myapp in the browser.
  Local:          http://localhost:3000
```

```
● ● ● localhost:3000

[ spinning React logo ]

Edit src/App.js and save to reload.

Learn React
```

Default React page opens in the browser

Press **Ctrl + C** in the terminal to stop the app before typing Git commands (or open a second Git Bash window).

If create-react-app gives errors (it is an older tool now), ask your lab in-charge, or use `npm create vite@latest myapp -- --template react` — the Git steps below stay the same.

Part 2 — Initialize Git and Push to GitHub

Step 1 — Initialize and commit

● ● ● Git Bash

```
$ git init
Reinitialized existing Git repository in C:/Users/You/myapp/.git/
$ git add .
$ git commit -m "Initial React app"
On branch main
nothing to commit, working tree clean
```

"Reinitialized" and "nothing to commit" are NOT errors! create-react-app already ran **git init** and made the first commit for you. Check it:

● ● ● Git Bash

```
$ git log --oneline
a1b2c3d (HEAD -> main) Initialize project using Create React App
```

Step 2 — Create an **empty** repo on GitHub named **myapp**

github.com → + → **New repository** → name **myapp** → **don't** tick README → **Create repository**

Step 3 — Connect and push (replace *username* with yours)

● ● ● Git Bash

```
$ git branch -M main
$ git remote add origin https://github.com/username/myapp.git
$ git push -u origin main
Writing objects: 100% (22/22), 180.50 KiB | 5.00 MiB/s, done.
To https://github.com/username/myapp.git
* [new branch]      main -> main
branch 'main' set up to track 'origin/main'.
```

- **git branch -M main** → makes sure your branch is called "main".
- If asked for a password → paste your **token**.

Part 3 — Use **.gitignore** to Exclude **node_modules**

Step 1 — Look inside **.gitignore** (create-react-app already made it)

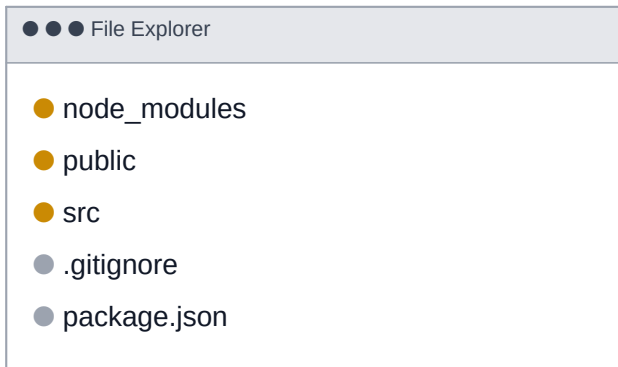
● ● ● Git Bash

```
$ cat .gitignore
# dependencies
/node_modules
...
# production
/build
```

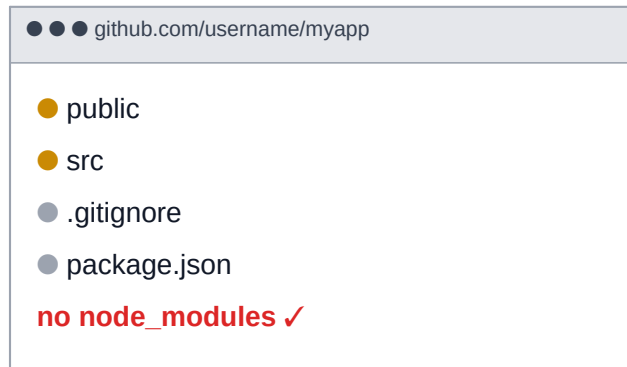
Part 3 — .gitignore (continued)

Line in .gitignore	Meaning
# dependencies	Lines starting with # are comments
/node_modules	Ignore the node_modules folder
/build	Ignore the build output folder

Step 2 — Proof it works — node_modules is on your computer but **not** on GitHub



Your computer (myapp folder)



GitHub repo page

Why ignore node_modules? It has thousands of files (100+ MB). Anyone can rebuild it by running `npm install`, because `package.json` lists everything needed.

Using Vite or a project without .gitignore? Create it yourself:



Part 4 — Branch 1: feature/navbar (merge with Pull Request)

Step 1 — Create the branch

```
● ● ● Git Bash

$ git checkout -b feature/navbar
Switched to a new branch 'feature/navbar'
```

Step 2 — Create a new file **src/Navbar.js**

```
function Navbar() {
  return (
    <nav>
      <h2>My App</h2>
      <p>Home | About | Contact</p>
    </nav>
  );
}

export default Navbar;
```

Step 3 — Replace everything in **src/App.js** with:

```
import Navbar from "../Navbar";

function App() {
  return (
    <div>
      <Navbar />
      <h1>Welcome to My React App</h1>
    </div>
  );
}

export default App;
```

Step 4 — Commit and push this branch to GitHub

```
● ● ● Git Bash

$ git add .
$ git commit -m "Add navbar component"
$ git push -u origin feature/navbar
* [new branch]      feature/navbar -> feature/navbar
```

Part 5 — Pull Request on GitHub (GUI)

Step	Do this on github.com/username/myapp
1	You'll see a yellow bar: "feature/navbar had recent pushes" → click Compare & pull request
2	Check: base: main ← compare: feature/navbar (see picture below)
3	Title: <i>Add navbar component</i> → click Create pull request
4	Wait for " No conflicts with base branch " → click Merge pull request
5	Click Confirm merge → you'll see " Pull request successfully merged " ✓
6	(Optional) Click Delete branch — the work is safe inside main now

● ● ● github.com/username/myapp/compare

Open a pull request

base: **main** ← compare: **feature/navbar** ✓ **Able to merge**

Add navbar component

Create pull request

GitHub — pull request page

Step 7 — Back on your computer — download the merged main

```
● ● ● Git Bash

$ git checkout main
Switched to branch 'main'
$ git pull
Updating a1b2c3d..b7c8d9e
Fast-forward
 src/App.js      | 10 ++++++--
 src/Navbar.js   | 10 ++++++++
```

Why git pull? The merge happened on **GitHub**, not on your computer. **git pull** brings that merged code down to you. Always pull before starting a new branch!

Part 6 — Branch 2: feature/form (merge with Git command)

Step 1 — Create the branch (from the updated main)

```
● ● ● Git Bash
$ git checkout -b feature/form
```

Step 2 — Create a new file **src/Form.js**

```
import { useState } from "react";

function Form() {
  const [name, setName] = useState("");

  return (
    <div>
      <input
        placeholder="Enter your name"
        onChange={(e) => setName(e.target.value)}
      />
      <p>Hello {name}</p>
    </div>
  );
}

export default Form;
```

Step 3 — In **src/App.js**, add the **2 lines** marked "<-- add this":

```
import Navbar from "../Navbar";
import Form from "../Form";          // <-- add this

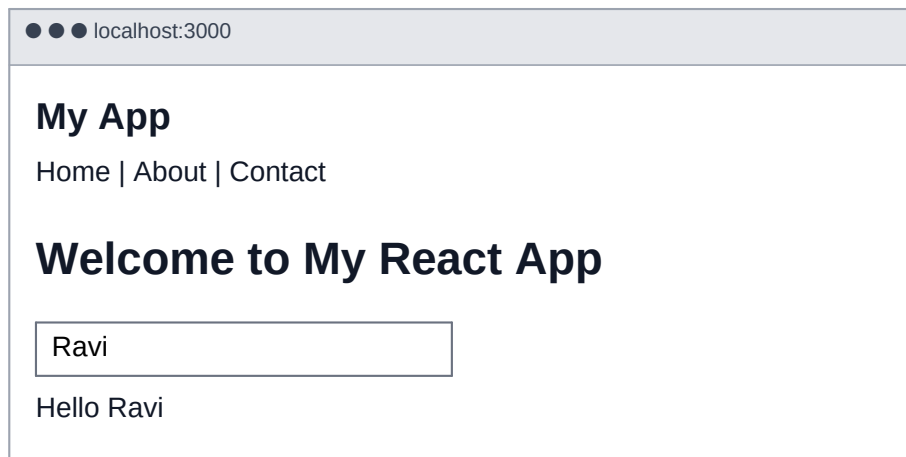
function App() {
  return (
    <div>
      <Navbar />
      <h1>Welcome to My React App</h1>
      <Form />                        {/* <-- add this */}
    </div>
  );
}

export default App;
```

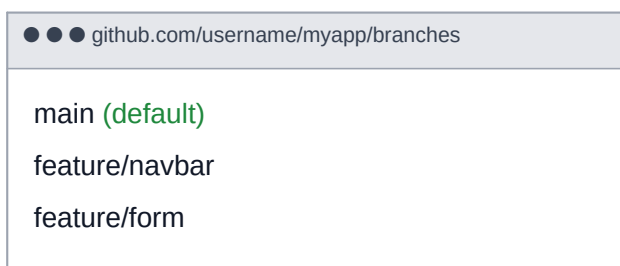
Step 4 — Commit, push the branch, then merge into main **on your computer**

```
● ● ● Git Bash
$ git add .
$ git commit -m "Add form component"
$ git push -u origin feature/form
$ git checkout main
$ git merge feature/form
Fast-forward
$ git push
```

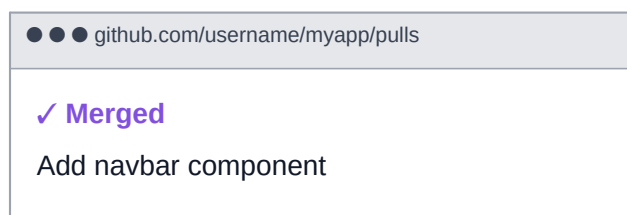
Final Output



npm start — app after both branches are merged



GitHub → Branches



GitHub → Pull requests → Closed

Merge vs Pull Request

	git merge (Part 6)	Pull Request (Part 5)
Where	Your computer (terminal)	GitHub website (buttons)
Review	No review	Team can review & comment first
After	git push to upload	git pull to download
Used in	Solo / small projects	Real teams & companies

RESULT

A React app was created with create-react-app and pushed to GitHub. node_modules was excluded using .gitignore. Two branches (feature/navbar and feature/form) were created and merged into main using a GitHub pull request and the git merge command respectively.